Why am I getting a different answer for these two python scripts?

investment = 10000
 for i in range(20):
yearly_interest = investment * .05
investment = investment + yearly_interest
 investment = round(investment, 2)
print(investment)

result is 26532.98

year = 0
investment = 10000
while year < 20:
yearly_interest = investment * .05
investment = investment + yearly_interest
 year += 1
investment = round(investment, 2)
print(investment)

result is 26532.97

You're seeing **two different final balances** because in the *while-loop version* you are **rounding inside the loop**, while in the *for-loop version* you round **only once at the end**.

That tiny difference compounds over 20 iterations.

The **while-loop script rounds every year**, introducing small truncation errors that accumulate.
The **for-loop script keeps full floating-point precision for all 20 years**, then rounds once.
That's why you get **26532.98** vs **26532.97**.

🔍 Breakdown of the difference

✓ Version 1 — *for-loop*

You do **no rounding** until the very end:

python
```
investment = 10000
for i in range(20):
    yearly_interest = investment * .05
    investment = investment + yearly_interest
investment = round(investment, 2)
print(investment)
```

This keeps full precision for all intermediate values. Final result: **26532.98**

✔ Version 2 — *while-loop*

You round **every iteration**:

python
```
year = 0
investment = 10000
while year < 20:
    yearly_interest = investment * .05
    investment = investment + yearly_interest
    year += 1
    investment = round(investment, 2)
print(investment)
```

Each year you chop off fractions of a cent. Those tiny losses accumulate over 20 years. Final result: **26532.97**

📌 Why rounding inside the loop changes the result

Each iteration loses up to **0.004999…** due to rounding. Over 20 years, that can add up to a cent or more.

This is normal in financial calculations — rounding frequency matters.

🛠 How to make both scripts match

Round **only once at the end**:

```
year = 0
investment = 10000
while year < 20:
    yearly_interest = investment * .05
    investment = investment + yearly_interest
    year += 1

investment = round(investment, 2)
print(investment)
```

This will give **26532.98**, matching the for-loop.